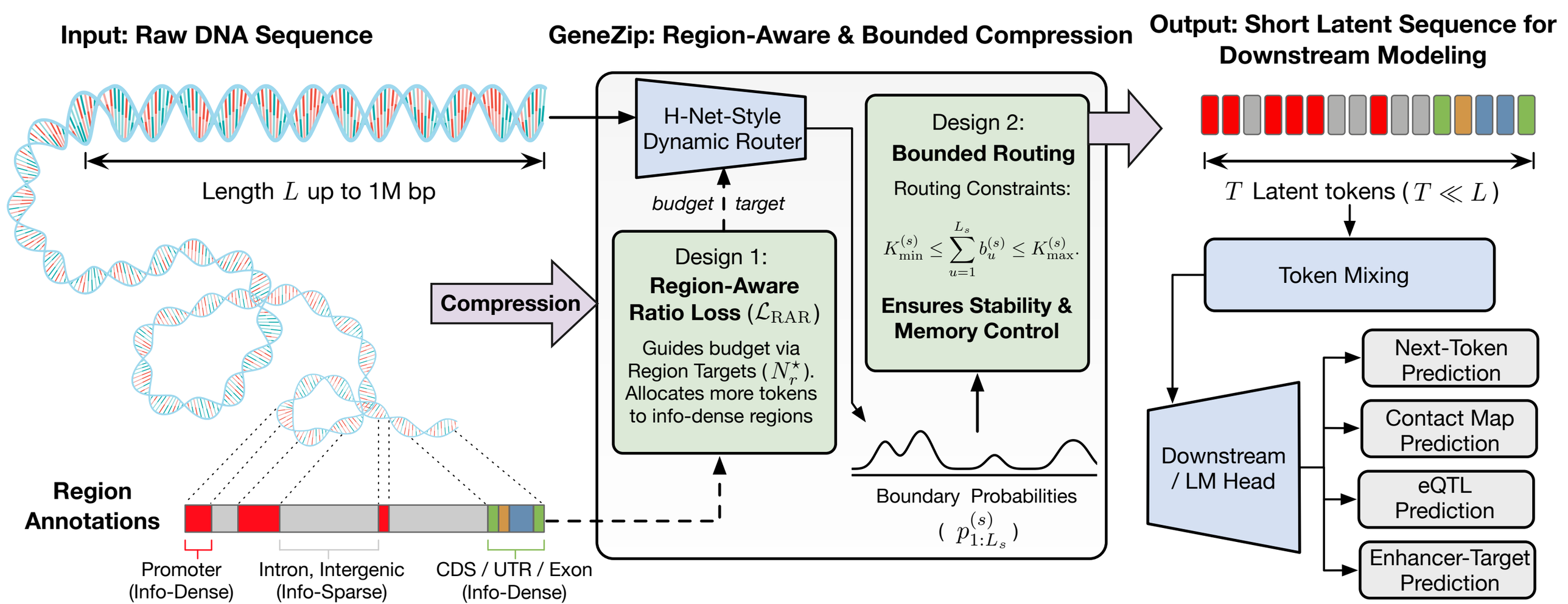
Input: Raw DNA Sequence
GeneZip: Region-Aware & Bounded Compression
Output: Short Latent Sequence for Downstream Modeling
Length L up to 1M bp
Compression
H-Net-Style Dynamic Router
budget
target
Design 1:
Region-Aware Ratio Loss ($\mathcal{L}_{\mathrm{RAR}}$)
Guides budget via Region Targets ($N_r^\star$). Allocates more tokens to info-dense regions
Design 2:
Bounded Routing
Routing Constraints:
$K_{\min}^{(s)} \le \sum_{u=1}^{L_s} b_u^{(s)} \le K_{\max}^{(s)}.$
Ensures Stability & Memory Control
Boundary Probabilities
($p_{1:L_s}^{(s)}$)
T Latent tokens ($T \ll L$)
Token Mixing
Downstream / LM Head
Next-Token Prediction
Contact Map Prediction
eQTL Prediction
Enhancer-Target Prediction
Region Annotations
Promoter (Info-Dense)
Intron, Intergenic (Info-Sparse)
CDS / UTR / Exon (Info-Dense)